

PE-EC702C — Neural Network & Fuzzy Logic Control

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Competitive Neural Networks

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Introduction to Competitive Learning	2
2	Kohonen Self-Organizing Maps (SOM)	2
2.1	Kohonen SOM Learning Algorithm	2
3	Learning Vector Quantization (LVQ)	3
3.1	LVQ Update Rules	3
4	Counter Propagation Networks (CPN)	3
4.1	CPN Architecture	3
4.2	CPN Training phases	4
5	Worked Examples	4

1. Introduction to Competitive Learning

Unlike feedforward networks that update weights based on output error signals, competitive learning networks are based on a winner-take-all strategy where nodes compete with each other to become active.

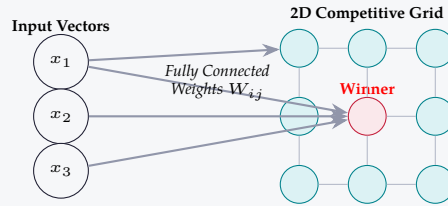
Competitive Learning

Competitive Learning is an unsupervised learning process where output neurons compete for activation. Only one neuron (the winning neuron) remains active at a time, and its weights are updated to become more similar to the input vector.

2. Kohonen Self-Organizing Maps (SOM)

Introduced by Teuvo Kohonen in 1982, the Self-Organizing Map (SOM) is an unsupervised learning network that maps high-dimensional input patterns to a low-dimensional (typically 1D or 2D) topological grid of neurons.

Kohonen Self-Organizing Map Architecture



2.1 Kohonen SOM Learning Algorithm

The training process involves four major steps:

1. **Initialization:** Initialize weight vectors $W_j = (w_{j1}, w_{j2}, \dots, w_{jn})^T$ for each grid node j with small random values. Set parameters: initial learning rate α_0 and initial neighborhood radius σ_0 .
2. **Similarity Matching (Finding the BMU):** For a given input vector $X = (x_1, x_2, \dots, x_n)^T$, compute the Euclidean distance to all weight vectors:

$$d_j(X) = \|X - W_j\| = \sqrt{\sum_{i=1}^n (x_i - w_{ji})^2} \quad (1)$$

Identify the winning neuron (Best Matching Unit, BMU) index c :

$$c = \arg \min_j d_j(X) \quad (2)$$

3. **Weight Updating:** Update weights of the BMU and all neurons within its neighborhood $h_{cj}(t)$:

$$W_j(t+1) = W_j(t) + \alpha(t)h_{cj}(t)[X - W_j(t)] \quad (3)$$

4. **Decay Parameters:** Decay the learning rate and the neighborhood size:

$$\alpha(t) = \alpha_0 \exp\left(-\frac{t}{\tau_1}\right), \quad \sigma(t) = \sigma_0 \exp\left(-\frac{t}{\tau_2}\right) \quad (4)$$

3. Learning Vector Quantization (LVQ)

Learning Vector Quantization (LVQ), introduced by Kohonen, is a supervised competitive learning algorithm used for pattern classification.

LVQ Architecture

An **LVQ network** consists of an input layer and a competitive layer containing codebook vectors (prototype vectors). Each codebook vector is assigned a specific target class.

3.1 LVQ Update Rules

Let w_c be the codebook vector closest to the input vector x (in terms of Euclidean distance). The weights are adjusted as follows:

LVQ Supervised Training updates

If the winning codebook vector w_c belongs to the **correct class** as the input x :

$$w_c(t+1) = w_c(t) + \alpha(t)[x - w_c(t)] \quad (5)$$

If the winning codebook vector w_c belongs to an **incorrect class** relative to input x :

$$w_c(t+1) = w_c(t) - \alpha(t)[x - w_c(t)] \quad (6)$$

All other codebook vectors remain unchanged:

$$w_j(t+1) = w_j(t) \quad \text{for } j \neq c \quad (7)$$

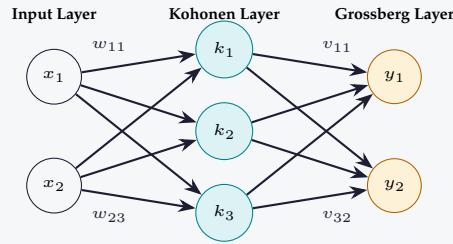
4. Counter Propagation Networks (CPN)

Developed by Robert Hecht-Nielsen in 1987, CPN combines a competitive unsupervised layer (Kohonen SOM) with a self-organizing outstar layer (Grossberg outstar). It acts as a fast lookup table to approximate continuous mathematical functions.

4.1 CPN Architecture

CPN contains three layers: Input layer, Kohonen layer (competitive), and Grossberg (Outstar) output layer.

Counter Propagation Network (CPN) Layout



4.2 CPN Training phases

1. **First Phase (Kohonen Training):** The input vector X is presented. The Kohonen layer acts as a competitive classifier. The winning neuron c is determined, and its weights W_c are updated using standard Kohonen rules.
2. **Second Phase (Grossberg Training):** The Kohonen layer outputs a binary vector where only the winning unit has activation 1, and all others are 0 ($z_j = \delta_{jc}$). The Grossberg weights V are updated to match the target vector Y :

$$v_{jk}(t+1) = v_{jk}(t) + \beta[y_k - v_{jk}(t)]z_j \quad (8)$$

For the winner c : $V_c(t+1) = V_c(t) + \beta[Y - V_c(t)]$.

5. Worked Examples

Kohonen SOM Winning Neuron and Weight Update

Problem: Given an input vector $X = \begin{bmatrix} 0.8 & 0.6 \end{bmatrix}^T$ and three Kohonen neurons with weight vectors:

$$W_1 = \begin{bmatrix} 0.9 & 0.1 \end{bmatrix}^T, \quad W_2 = \begin{bmatrix} 0.7 & 0.5 \end{bmatrix}^T, \quad W_3 = \begin{bmatrix} 0.2 & 0.8 \end{bmatrix}^T$$

- (a) Identify the winning neuron (BMU) using Euclidean distance.
- (b) Update the weight vector of the winning neuron using learning rate $\alpha = 0.5$.

Solution:

(a) **Similarity Matching:** Compute the squared Euclidean distance $D_j = \|X - W_j\|^2$ for each neuron:

- For W_1 :

$$D_1 = (0.8 - 0.9)^2 + (0.6 - 0.1)^2 = (-0.1)^2 + (0.5)^2 = 0.01 + 0.25 = 0.26$$

- For W_2 :

$$D_2 = (0.8 - 0.7)^2 + (0.6 - 0.5)^2 = (0.1)^2 + (0.1)^2 = 0.01 + 0.01 = 0.02$$

- For W_3 :

$$D_3 = (0.8 - 0.2)^2 + (0.6 - 0.8)^2 = (0.6)^2 + (-0.2)^2 = 0.36 + 0.04 = 0.40$$

Since $D_2 = 0.02$ is the minimum, **Neuron 2** (W_2) is the winning neuron (BMU).

(b) Weight Update: Update the weights of the winning neuron W_2 :

$$\begin{aligned} W_2(new) &= W_2(old) + \alpha[X - W_2(old)] \\ &= \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix} + 0.5 \left(\begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} - \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix} + 0.5 \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 0.05 \\ 0.05 \end{bmatrix} = \begin{bmatrix} 0.75 \\ 0.55 \end{bmatrix} \end{aligned}$$

The updated weight vector is $W_2 = \begin{bmatrix} 0.75 & 0.55 \end{bmatrix}^T$.

LVQ Classifier Updates

Problem: Given an input vector $x = \begin{bmatrix} 0.1 & 0.9 \end{bmatrix}$ belonging to Class 1, and the closest codebook vector $w_c = \begin{bmatrix} 0.2 & 0.7 \end{bmatrix}$. Calculate the updated codebook vector for:

- Class of w_c is Class 1 (correct class)
- Class of w_c is Class 2 (incorrect class)

Assume learning rate $\alpha = 0.2$.

Solution:

(a) Correct Class Update (Push towards input):

$$\begin{aligned} w_c(new) &= w_c(old) + \alpha[x - w_c(old)] \\ &= \begin{bmatrix} 0.2 \\ 0.7 \end{bmatrix} + 0.2 \left(\begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix} - \begin{bmatrix} 0.2 \\ 0.7 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0.2 \\ 0.7 \end{bmatrix} + 0.2 \begin{bmatrix} -0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.2 - 0.02 \\ 0.7 + 0.04 \end{bmatrix} = \begin{bmatrix} 0.18 \\ 0.74 \end{bmatrix} \end{aligned}$$

(b) Incorrect Class Update (Pull away from input):

$$\begin{aligned} w_c(new) &= w_c(old) - \alpha[x - w_c(old)] \\ &= \begin{bmatrix} 0.2 \\ 0.7 \end{bmatrix} - 0.2 \left(\begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix} - \begin{bmatrix} 0.2 \\ 0.7 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0.2 \\ 0.7 \end{bmatrix} - 0.2 \begin{bmatrix} -0.1 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.2 + 0.02 \\ 0.7 - 0.04 \end{bmatrix} = \begin{bmatrix} 0.22 \\ 0.66 \end{bmatrix} \end{aligned}$$

Quick Revision — Module II

Competitive Neural Networks | PE-EC702C

Key Formulas at a Glance

Euclidean Distance in SOM	$d_j(X) = \sqrt{\sum_{i=1}^n (x_i - w_{ji})^2}$
Winning Neuron (BMU) Index	$c = \arg \min_j d_j(X)$
Kohonen Weight Update	$W_j(t+1) = W_j(t) + \alpha(t)h_{cj}(t)[X - W_j(t)]$
LVQ Correct Update	$w_c(t+1) = w_c(t) + \alpha(t)[x - w_c(t)]$
LVQ Incorrect Update	$w_c(t+1) = w_c(t) - \alpha(t)[x - w_c(t)]$
Grossberg outstar Update	$V_c(t+1) = V_c(t) + \beta[Y - V_c(t)]$

One-Line Definitions

- **Kohonen SOM:** An unsupervised clustering network that creates a topology-preserving mapping of high-dimensional input features.
- **Best Matching Unit (BMU):** The neuron in a competitive layer whose weight vector is closest to the current input vector.
- **LVQ:** A supervised competitive network that uses prototype vectors to map inputs to specific classification boundaries.
- **CPN:** A hybrid network that maps input patterns to target vectors rapidly by combining a Kohonen layer with a Grossberg layer.
- **Grossberg Outstar:** A learning unit designed to output a specific analog target pattern when triggered by a Kohonen node.

Frequently Asked / Viva Questions

1. **What is the primary difference between Perceptron learning and Kohonen learning?**
Perceptron learning is supervised and minimizes output error feedback. Kohonen learning is unsupervised and uses similarity metrics to cluster features.
2. **How does the neighborhood function $h_{cj}(t)$ change during SOM training?**
It starts wide to organize the global structure of the map, and then shrinks exponentially over time to allow fine-tuning of local weights.
3. **Why is LVQ considered a supervised learning algorithm while SOM is unsupervised?**
SOM groups data based solely on input similarities without class labels. LVQ adjusts its codebook vectors based on class labels to optimize classification boundaries.
4. **Explain the role of Grossberg outstar layer in Counter Propagation Networks.**

Frequently Asked / Viva Questions (contd.)

The Kohonen layer maps inputs to a single active winning unit. The Grossberg layer acts as a lookup table, outputting a stored target vector associated with that unit.

5. What is the significance of the vigilance parameter in competitive networks?

It controls the sensitivity of node matching. In networks like ART, it determines if a new cluster must be created or if the input fits an existing cluster.

6. Why does CPN train faster than Backpropagation networks?

CPN updates only the winning Kohonen neuron and the corresponding Grossberg weights, avoiding expensive error propagation across multiple layers.

7. What is the purpose of lateral inhibition in competitive networks?

It allows active neurons to suppress the activation of neighboring competitive neurons, ensuring that only one winning neuron (the BMU) remains active.

8. How does LVQ-2 differ from standard LVQ-1?

LVQ-1 updates only the single closest codebook vector. LVQ-2 updates the two closest vectors simultaneously if the input falls near the decision boundary.

9. What is meant by a topology-preserving mapping in SOM?

It means that input vectors that are close in the high-dimensional feature space are mapped to neurons that are close to each other on the 2D output grid.

10. What is the codebook vector in Learning Vector Quantization?

It is a prototype vector representing a class cluster center in the input space.

Comparison of Competitive Networks

Attribute	Kohonen SOM	LVQ	CPN
Learning Type	Unsupervised	Supervised	Hybrid (Supervised + Unsupervised)
Primary Application	Dimensionality reduction	Pattern classification	Function approximation
Output Layer Grid	1D or 2D grid structure	1D array of vectors	1D array of output nodes
Weight Updates	Updates winner + neighbors	Updates winner only	Updates winner only